

23.1

Discovolution (1)

23.1.1 Discovolution: UGC NET CSE | December 2015 | Part 3 | Question: 51



Blind image deconvolution is

- A. Combination of blur identification and image restoration B. Combination of segmentation and classification
C. Combination of blur and non-blur image D. None of the above

ugcnetcse-dec2015-paper3 digital-image-processing discovolution

Answer key

23.2

Entropy (2)

23.2.1 Entropy: UGC NET CSE | December 2014 | Part 3 | Question: 65



Given a simple image of size 10×10 whose histogram models the symbol probabilities and is given by

P_1	P_2	P_3	P_4
a	b	c	d

The first order estimate of image entropy is maximum when

- A. $a=0, b=0, c=0, d=1$ B. $a=\frac{1}{2}, b=\frac{1}{2}, c=0, d=0$
C. $a=\frac{1}{3}, b=\frac{1}{3}, c=\frac{1}{3}, d=0$ D. $a=\frac{1}{4}, b=\frac{1}{4}, c=\frac{1}{4}, d=\frac{1}{4}$

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Answer key

23.2.2 Entropy: UGC NET CSE | December 2015 | Part 3 | Question: 49



Consider the conditional entropy and mutual information for the binary symmetric channel. The input source has alphabet $X = \{0, 1\}$ and associated probabilities $\frac{1}{2}, \frac{1}{2}$. The channel matrix is $\begin{pmatrix} 1-p & p \\ p & 1-p \end{pmatrix}$ where p is the transition probability. Then the conditional entropy is given by:

- A. 1 B. $-p \log(p) - (1-p) \log(1-p)$
C. $1 + p \log(p) + (1-p) \log(1-p)$ D. 0

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Answer key

23.3

Fourier Transform (1)

23.3.1 Fourier Transform: UGC NET CSE | December 2012 | Part 3 | Question: 64



If the Fourier transform of the function $f(x, y)$ is $F(m, n)$, then the Fourier transform of the function $f(2x, 2y)$ is

- A. $\frac{1}{4} F\left(\frac{m}{2}, \frac{n}{2}\right)$ B. $\frac{1}{4} F(2m, 2n)$ C. $\frac{1}{4} F(m, n)$ D. $\frac{1}{4} F\left(\frac{m}{4}, \frac{n}{4}\right)$

ugcnetcse-dec2012-paper3 fourier-transform digital-image-processing

23.4

Image Compression (4)

23.4.1 Image Compression: UGC NET CSE | July 2016 | Part 3 | Question: 60



Which of the following is used for the boundary representation of an image object?

- A. Quad Tree B. Projections C. Run length coding D. Chain codes

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Answer key

23.4.2 Image Compression: UGC NET CSE | June 2012 | Part 3 | Question: 10



In an image compression system 16384 bits are used to represent 256×256 image with 256 gray levels. What is the compression ratio for this system?

- A. 1 B. 2 C. 4 D. 8

ugcnetcse-june2012-paper3 image-compression

Answer key

23.4.3 Image Compression: UGC NET CSE | June 2012 | Part 3 | Question: 19



The quantiser in an image-compression system is a

- A. lossy element which exploits the psychovisual redundancy
B. lossless element which exploits the psychovisual redundancy
C. lossy element which exploits the statistical redundancy
D. lossless element which exploits the statistical redundancy

ugcnetcse-june2012-paper3 image-compression

Answer key

23.4.4 Image Compression: UGC NET CSE | June 2014 | Part 3 | Question: 55



Given an image of size 1024×1024 pixels in which intensity of each pixel is an 8-bit quality. It requires _____ of storage space if the image is not compressed.

- A. One Terabyte B. One Megabyte
C. 8 Megabytes D. 8 Terabytes

ugcnetjune2014iii image-compression computer-graphics

Answer key

Answer Keys

23.1.1	D
23.4.2	X

23.2.1	D
23.4.3	A

23.2.2	D
23.4.4	B

23.3.1	A
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23.4.1	D
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